DEPI | DATA ANALYSIS TRACK | GRADUATION PROJECT

# Project Planning & Management

Vulnerability-to-Exploitation
Cybersecurity Analytics (2016-2026)

Project proposal | Project plan | Roles and tasks | Risk plan | KPIs


PROJECT TEAM

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GitHub: github.com/nhahub/NHA-5-223

# Project at a Glance

One page that summarizes what the project is, where its data comes from, and how the work flows from collection to recommendations.

## From Vulnerability to Exploit

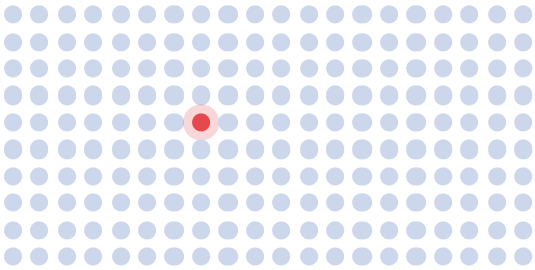
Which vulnerabilities actually get exploited, how fast, and why? | Cybersecurity Analytics 2016-2026

DEPI DATA ANALYSIS

| 10 | 5 | ~270K | <0.5% | 7 | 4 |
|---|---|---|---|---|---|
| YEARS OF DATA | REAL PUBLIC SOURCES | CVES (ESTIMATE) | CONFIRMED EXPLOITED | ANALYSIS QUESTIONS | DASHBOARD TOOLS |

### THE NEEDLE IN THE HAYSTACK

Each dot is 0.5% of published CVEs. Only about one dot in this grid is ever exploited.



CVSS tells us how **severe** a flaw could be, not whether attackers will use it. Patching by severity alone spends effort on thousands of flaws that are never exploited.

### FIVE SOURCES, ONE KEY

All sources are joined on the CVE ID into one dataset.

| | | |
|---|---|---|
| NVD | **NIST National Vulnerability Database** — CVE records, CVSS, CWE, vendor, product | BASE TABLE |
| KEV | **CISA Known Exploited Vulns** — Confirmed exploitation + date added | LABEL |
| EPSS | **FIRST.org EPSS** — Daily exploitation probability (0-1) | BASELINE |
| CWE | **MITRE CWE** — Names and categories of weaknesses | ENRICHMENT |
| EDB | **Exploit-DB** — Public exploit code mapped to CVEs | FEATURE |

1. **Collect** — Download NVD feeds 2016-2026, KEV, EPSS, CWE, Exploit-DB. Log source and date.
2. **Clean & Integrate** — Standardize, remove rejected CVEs, left-join on CVE ID, document gaps.
3. **Store & Validate** — Load into SQL, run analytical queries, reconcile counts.
4. **Analyze** — Answer 7 questions: volume, exploit rate, CVSS vs EPSS, time-to-KEV.
5. **Visualize** — Dashboards in Python, Excel, Power BI and Tableau with the same numbers.
6. **Recommend** — Report and presentation with evidence-based patching priorities.

**Impact: from severity-based to evidence-based patching**

Team of 6 | 4 weeks | 100% real, publicly verifiable data

Python SQL Excel Power BI Tableau

# 1. Project Proposal

## 1.1 Overview

Every year, tens of thousands of new software vulnerabilities (CVEs) are published, yet only a very small fraction is ever exploited in real attacks. Security teams usually prioritize patching by CVSS severity score alone, which measures theoretical severity rather than real-world risk. As a result, effort is often spent on vulnerabilities attackers never use, while truly dangerous ones wait in the queue.

This project analyzes **ten years of real vulnerability data (2016 to 2026)** from five authoritative public sources to answer one question: **which vulnerabilities actually get exploited, how fast, and why?** The findings are delivered as dashboards in Python, Excel, Power BI, and Tableau, backed by a SQL database and a final report.

## 1.2 Problem Statement

- The number of published CVEs grows every year, so patching everything is unrealistic.
- CVSS reflects potential severity, not actual exploitation.
- Security teams need evidence-based prioritization to decide what to patch first.

## 1.3 Objectives

| No. | Objective | How it will be measured |
|---|---|---|
| O1 | Build one clean, unified dataset by integrating five authoritative sources | Five sources joined on CVE ID with no duplicate IDs |
| O2 | Analyze ten-year trends in volume, severity, and exploitation | Yearly trend analysis completed and documented |
| O3 | Measure how well CVSS severity reflects actual exploitation | Exploitation rate per CVSS band calculated |
| O4 | Identify the vendors, products, and CWE types most associated with exploited vulnerabilities | Ranked lists with supporting counts |
| O5 | Measure the time between publication and confirmed exploitation | Days from publication to KEV listing, by year |
| O6 | Compare CVSS and EPSS and study where they disagree | Disagreement cases identified and described |
| O7 | Present findings in interactive dashboards with actionable insights | Four dashboards with consistent numbers |

## 1.4 Scope

| In scope | Out of scope |
|---|---|
| ■ Collect real data from the five public sources and log each one<br>■ Build one cleaned, unified dataset linked by CVE ID<br>■ Store it in a SQL database with analytical queries<br>■ Answer the seven diagnostic questions<br>■ Build dashboards in Python, Excel, Power BI, and Tableau<br>■ Deliver a final report, a presentation, and a documented repository | ■ Machine learning and predictive modeling<br>■ Live data pipelines or real-time monitoring<br>■ Sources beyond the five listed (CVE List V5 only to fill NVD gaps) |

## 1.5 Data Sources

| Source | Provider | Content | Role |
|---|---|---|---|
| NVD (2016-2026) | NIST | CVE records, CVSS scores and vectors, CWE IDs, vendors and products (CPE), dates | Base table |
| CISA KEV | CISA | Vulnerabilities confirmed as exploited in the wild, with date added | Exploitation label |
| EPSS | FIRST.org | Daily probability score (0-1) of exploitation | Comparison baseline |
| MITRE CWE | MITRE | Names and categories of weakness types | Enrichment |
| Exploit-DB | OffSec | Public exploit code mapped to CVE IDs | Public exploit flag |

## 1.6 Expected Impact

- Moves patch prioritization from severity-based to evidence-based.
- Quantifies how reliable CVSS is as a prioritization signal.
- Demonstrates an end-to-end data analysis workflow on large-scale, 100% real, publicly verifiable cybersecurity data.

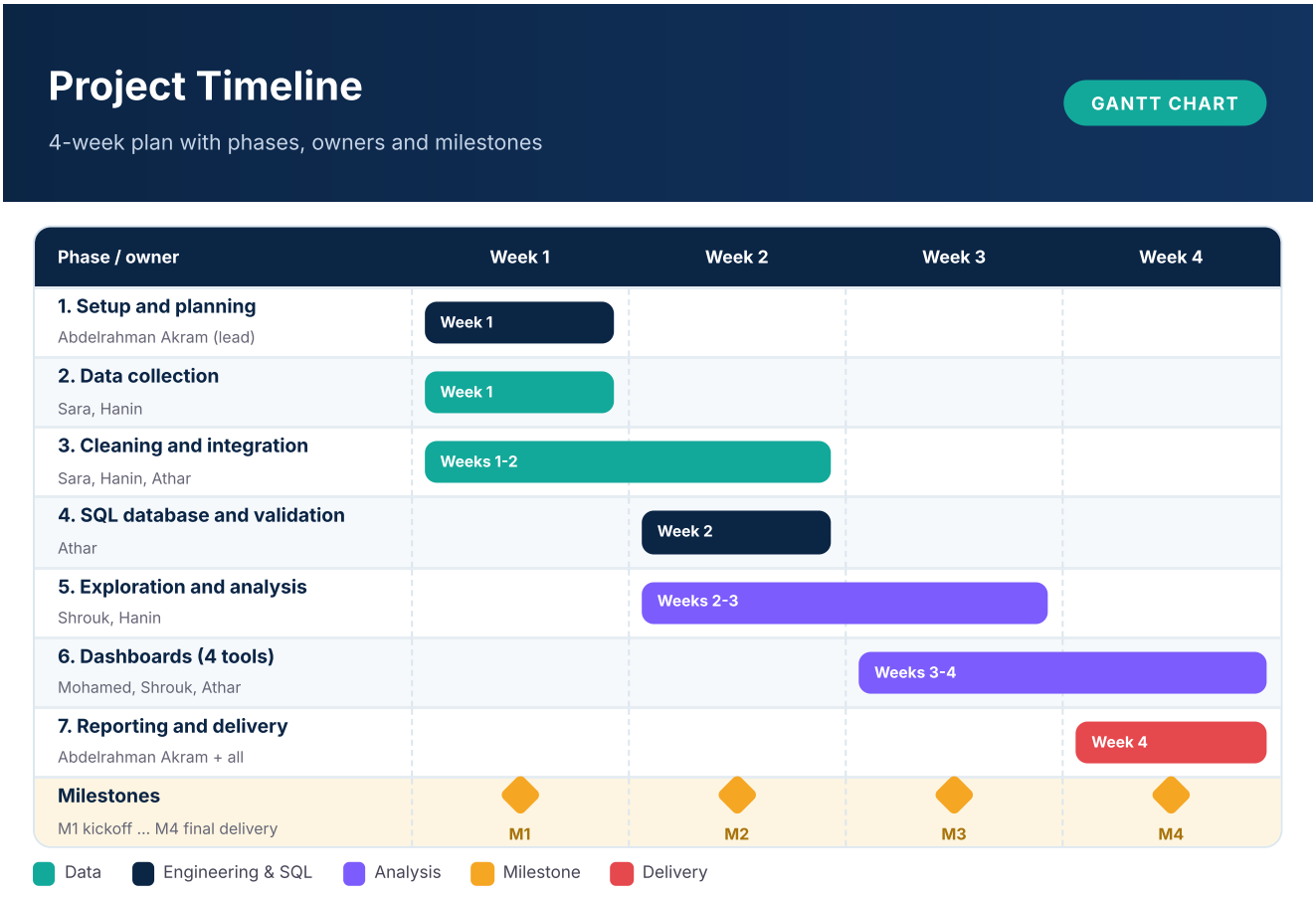
# 2. Project Plan

The project runs for **4 weeks** in seven overlapping phases. The duration is a proposal and will be adjusted to the official deadline.

## 2.1 Phases

| Phase | Weeks | Main activities | Output |
|---|---|---|---|
| **1. Setup and planning** | 1 | Finalize scope and roles; create the repository and task board; request the NVD API key; agree on Git rules | Approved plan, board, repository |
| **2. Data collection** | 1 | Download NVD feeds, KEV, EPSS, CWE, and Exploit-DB; record source, version, and date of every file | Raw data, data source log |
| **3. Cleaning and integration** | 1-2 | Clean and standardize each source; join all sources on CVE ID; document missing values | Unified dataset, data dictionary |
| **4. SQL database and validation** | 2 | Load the dataset into a relational database; write analytical queries; reconcile counts | SQL schema, query library |
| **5. Exploration and analysis** | 2-3 | Answer the seven analysis questions and document findings with charts | Findings notebook, insight list |
| **6. Dashboards** | 3-4 | Build the Python, Excel, Power BI, and Tableau dashboards from the same validated data | Four reviewed dashboards |
| **7. Reporting and delivery** | 4 | Write the final report, prepare the presentation, clean up the repository | Report, slides, final repository |

## 2.2 Timeline (Gantt Chart)



## 2.3 Deliverables

| Deliverable | Due |
|---|---|
| Project planning document (this file) | Week 1 |
| Data source log (source, version, download date) | Week 1 |
| Unified dataset and data dictionary | Week 2 |
| Missing-values report | Week 2 |
| SQL schema and query library | Week 2 |
| Analysis notebook and insight list | Week 3 |
| Python dashboard | Week 3 |
| Excel dashboard | Week 3 |
| Power BI dashboard | Week 4 |
| Tableau dashboard | Week 4 |
| Final report | Week 4 |
| Presentation | Week 4 |
| Documented GitHub repository | Week 4 |

## 2.4 Tools and data resources

| Resource | Use in the project |
|---|---|
| GitHub | Code, documentation, task board, pull requests |
| Python (pandas, matplotlib, seaborn) | Download, parse, clean, merge, analyze, Python dashboard |
| SQL database | Store the unified dataset and run analytical queries |
| Excel | Data checks, pivot tables, interactive dashboard |
| Power BI Desktop | Data model, DAX measures, interactive dashboard |
| Tableau | Complementary dashboards and storytelling |
| NVD API key | Faster, higher-limit downloads from NVD |

# 3. Task Assignment and Roles

## 3.1 Team Roles

| Member | Role | Responsibilities |
|---|---|---|
| **Abdelrahman Akram Ibrahim Mohammed** Team Leader | Data Collection and Tableau Dashboard | Plans and tracks tasks; manages the repository; reviews pull requests; collects the data; maintains documentation and data sources; develops the Tableau dashboard. |
| **Sara Ahmed Ali Ahmed** | Exploration, Cleaning and Python Dashboard | Explores the dataset to find patterns and trends; cleans and prepares the data; performs analysis; develops the Python dashboard. |
| **Hanin Abdo** | Power BI Analyst | Builds the Power BI dashboard and visualizations. Creates interactive filters and KPI cards. |
| **Athar Ashraf** | Data Quality and SQL Analyst | Documents data quality and missing values; builds the SQL database and queries; validates counts. |
| **Shrouk Muhamed** | Analyst (Excel) | Leads the exploratory and diagnostic analysis; builds summary tables and the Excel dashboard; co-leads the presentation. |
| **Mohamed Elsayed Abd Elrauf Aly** | ——— | ——— |

## 3.2 Ways of Working

- Tasks live on a board with the columns To Do, In Progress, In Review, and Done; every task has an owner and a due week.
- Each task is developed in its own Git branch and merged through a pull request reviewed by another member.
- Large raw data files stay out of Git; the repository keeps source links and the data source log instead.
- Weekly planning meeting (all members) to review progress and blockers, plus a short check-in twice a week and a group chat for daily questions.

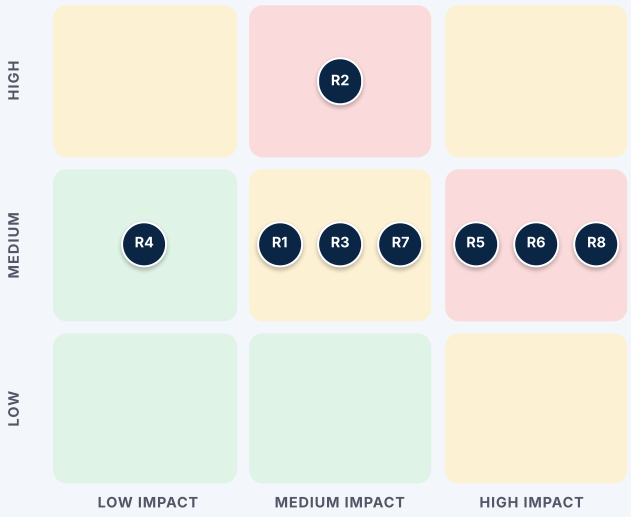
# 4. Risk Assessment and Mitigation Plan

Risks are rated by likelihood and impact (Low, Medium, High). A risk with a score of 6 or more is treated as high priority and is reviewed in every weekly meeting.

## Risk Heat Map

Likelihood (vertical) vs impact (horizontal) of the eight main project risks

RISK ASSESSMENT




**RISK REGISTER**

| | | |
|---|---|---|
| R1 | NVD API limits or slow downloads | MEDIUM |
| R2 | Missing or inconsistent data across sources | HIGH |
| R3 | Large dataset slows Excel or Power BI | MEDIUM |
| R4 | Source links or file formats change | LOW |
| R5 | A team member becomes unavailable | HIGH |
| R6 | Different numbers across dashboards | HIGH |
| R7 | Scope creep | MEDIUM |
| R8 | A phase runs late | HIGH |

## Risk Register

| Risk | Likelihood | Impact | Mitigation (prevent) | Contingency (if it happens) |
|---|---|---|---|---|
| NVD API limits or slow downloads | Medium | Medium | Request the free API key in week 1; use the annual feed files; start collection immediately | Use cached downloads; download in smaller batches |
| Missing or inconsistent data across sources | High | Medium | Document every gap; separate "not applicable" from "data unavailable"; review at M2 | Use CVE List V5 to fill NVD gaps; restrict the affected analysis to complete records |
| Large dataset slows Excel or Power BI | Medium | Medium | Load only the required columns; use SQL aggregations; test performance early | Pre-aggregate by year and vendor; use a filtered extract for Excel |
| Source links or file formats change | Medium | Low | Record source, version, and download date; keep copies of the raw files in shared storage | Switch to the source's alternative download page or API |
| Different numbers across dashboards | Medium | High | Use one validated dataset as the single source; check dashboard totals against SQL results | Freeze the dataset version; re-check all dashboards before M3 |
| A phase runs late | Medium | High | Track progress every few days; use milestone criteria; keep a buffer in week 4 | Cut optional work first; add members to the critical task |

# 5. Key Performance Indicators (KPIs)

The KPIs measure data quality, analytical output, dashboard quality, and team delivery. They replace generic software metrics with ones that fit a data analysis project: response time becomes dashboard response time, and user adoption becomes the usability rating given by reviewers. System uptime does not apply because the project has no live service.

## Success Scorecard

Target values the team will track every week

KPIs

| | | | |
|---|---|---|---|
| DATA<br>**5 / 5**<br>Sources collected, versioned and logged | DATA<br>**≥ 95%**<br>Completeness of publish date and CVSS score | DATA<br>**0**<br>Duplicate CVE IDs after all joins | DATA<br>**100%**<br>SQL key totals reconciled with the dataset |
| ANALYSIS<br>**7 / 7**<br>Diagnostic questions answered with evidence | DASHBOARDS<br>**4 / 4**<br>Dashboards showing identical headline numbers | PERFORMANCE<br>**< 5 s**<br>Response time of a dashboard filter on a laptop | ADOPTION<br>**≥ 4 / 5**<br>Usability rating from non-technical reviewers |
| DELIVERY<br>**≥ 3 / 4**<br>Milestones completed on time | DELIVERY<br>**≥ 90%**<br>Tasks closed by the end of their phase | TEAM<br>**100%**<br>Pull requests reviewed by a second member | TEAM<br>**≥ 90%**<br>Attendance at weekly planning meetings |

### Assumptions and Dependencies

- The public data sources stay available during the project.
- Power BI Desktop runs only on Windows, so the BI developer needs a Windows machine.
- Tableau Public or a student license is available to the team.
- The weekly time commitment of each member is agreed at the kickoff meeting.
- The final deadline is set by DEPI; this plan will be adjusted once it is confirmed.